

LOT2 — Review of Analytical Methodology

Scope. An independent check of the LOT2 pipeline (main.R and the scripts it sources) against its own outputs, asking whether the strategy, parameters and products are fit for the stated goal: characterising fungal endophyte diversity in this ecosystem. Every figure quoted below was recomputed from the current input files.

Inputs reviewed. LOT2_samples.xlsx (650 isolates), LOT2_pooled_counts.xlsx (285 rows), and all 200+ generated tables and plots.

Verdict

The pipeline is well engineered and the multivariate core is sound: the **substrate effect is real** and survives every reasonable challenge I applied, including adjustment for sampling depth. Reproducibility is good — one command regenerates everything.

But three results currently in the report should not be published as they stand: the **pie and abundance charts are contaminated by a spreadsheet totals row**, the **indicator species do not survive multiple-testing correction**, and the **host comparison is pseudoreplicated**. Separately, the study’s headline question — how diverse is this system — is answered with descriptive statistics that omit the standard estimators, even though the data support them.

The single largest missed opportunity: all 650 ITS sequences are sitting unused in the workbook.

1. What works well

- **Substrate is a genuine signal.** PERMANOVA $R^2 = 0.19$ ($p = 0.001$) at ITS level, 0.40 at genus level. It holds after adjusting for sampling depth (marginal $R^2 = 0.27$, $p = 0.001$), and within zone 6 alone, so it is not an artefact of sampling height.
- **Rank-by-rank repetition** (Section D) is a genuinely good design choice. It is what revealed that the ITS-level ordinations are unusable while the genus-level ones are fine.
- **Incertae sedis handled per rank** rather than globally — correct, and better than the common practice of dropping unresolved isolates entirely.
- **Orientation conclusion is correctly argued.** The negative result is properly defended with marginal tests that dissolve the two confounded positives.

2. Problems, ranked

P1 — CRITICAL: a totals row is plotted as a taxon

Row 286 of LOT2_pooled_counts.xlsx has no culture code and no taxon name, but carries the column sums: 219 / 264 / 167. parse_LOT2.R reads it as a genotype. Downstream, plot_pie.R and plot_abundance_pooled.R bin it as *Incertae sedis*.

Chart	<i>Incertae sedis</i> as plotted	Actual
Pie by phylum	50% / 50% / 51%	0% / 0% / 2%
Pie by genus	56% / 60% / 60%	11% / 20% / 20%

The README then interprets this slice as “how much of each community remains unidentified at that rank” — off by a factor of 25 at phylum level. It also explains why the two input files disagree by exactly $2\times$ in total (438/528/334 vs 219/264/167).

The multivariate analyses are **unaffected** (they read the sample-level file), and the diversity indices escaped because the row has no taxonomy and is filtered with the *Incertae sedis* bin. Only the pie and abundance charts are wrong.

Fix: drop rows with no culture code and no taxon name at parse time, and assert that pooled totals equal sample-level totals.

P2 — CRITICAL: no indicator species survives correction

multipatt p-values are filtered at raw $p \leq 0.05$ with no correction, across 238–283 taxa per analysis. Recomputing with Benjamini–Hochberg:

Analysis	Taxa tested	Raw $p \leq 0.05$	BH $q \leq 0.05$
Substrate	283	21	0
Sun aspect	238	4	0
Orientation	242	3	0

At 283 tests, ~14 hits at $p \leq 0.05$ are expected by chance alone. The 21 reported for substrate are not distinguishable from noise. Across all analyses the pipeline currently reports **142 “significant” indicator taxa**; none would survive.

Fix: apply BH-FDR within each analysis and report q-values. Expect to report zero indicators — which is an honest and interesting result given the singleton-dominated community, not a failure.

P3 — MAJOR: the host comparison is pseudoreplicated

As recorded, there is **no tree-level replication**. All 264 *Ficus* leaf isolates share one collection code (LOT02-67508) and the sheet’s Tree column reads “Ficus” throughout. No tree identifier exists anywhere in the data.

- *Ficus* leaves vs *Ficus* wood is a **within-tree tissue contrast** — valid, but describing one tree.
- *Ficus* leaves vs Lauraceae leaves is a **host contrast with $n = 1$ tree per host**. PERMANOVA treats 20 sampling units as independent replicates of “host”, so the p-value tests whether these two *individual trees* differ, not whether the two host species do.

This does not invalidate the numbers; it bounds what they mean. The README’s conclusion that “host identity adds a secondary but significant effect” is not supported at the species level.

Fix: state the inferential unit explicitly, and rephrase host claims as descriptive of the sampled individuals. If other trees were sampled, add the tree ID and use it as a stratum.

P4 — MAJOR: alpha diversity is computed with $n = 1$ per substrate

Section A derives all indices from the pooled file, giving **one value per substrate and no error, no test**. Statements like “the richest community is *Ficus* leaves” therefore have no statistical support.

The sample-level file already supports a proper test. Per sampling unit:

Substrate	Units	Mean H'	SD
<i>Ficus</i> leaves	10	2.92	0.16
<i>Ficus</i> wood	13	1.99	0.49
Lauraceae leaves	10	2.46	1.05

Kruskal–Wallis: **Shannon $p = 0.00049$, richness $p = 0.00037$** . The result the report wants is available and significant — it just is not being computed.

Fix: compute alpha diversity per sampling unit and test across substrates.

P5 — MAJOR: sampling completeness is never quantified

The README says rarefaction curves “have not saturated” but never estimates how far off they are. They are far off:

Substrate	Observed S	Chao1 ($\pm$ SE)	Recovered	Good's coverage	Singletons
Ficus leaves	135	302 $\pm$ 50	45%	65%	92
Ficus wood	82	206 $\pm$ 49	40%	67%	55
Lauraceae leaves	105	186 $\pm$ 28	56%	71%	63

Less than half the estimated richness was recovered, and about a third of isolates belong to genotypes seen exactly once. For a study whose goal is characterising diversity, these are the headline numbers and they are absent.

Fix: report Chao1/ACE with SEs and Good's coverage; compare diversity at equal *coverage* rather than equal *count*.

P6 – MAJOR: sampling depth is confounded with substrate

Ficus wood units yielded roughly half as many isolates as leaf units (9.6 vs 20.2 and 18.8 per unit at genus level). Depth is itself a significant predictor of composition:

- community ~ depth: $R^2 = 0.216$, $p = 0.001$ (genus level)
- marginal: depth $R^2 = 0.069$ ($p = 0.003$), substrate $R^2 = 0.266$ ($p = 0.001$)

Substrate survives, which is the important point — but part of the raw substrate effect is a depth effect, and this is neither reported nor adjusted for.

Fix: include depth as a covariate in the reported model, or standardise by coverage before comparing.

P7 – MAJOR: dispersion heterogeneity at the ranks the report relies on

At genus level — the ranks the report (correctly) recommends as the readable ones — **PERMDISP $p = 0.004$** , with Ficus wood far more internally variable (mean distance to centroid 0.538) than either leaf substrate (0.288, 0.322). PERMANOVA cannot distinguish “different centroids” from “different spread”, so the genus-level effect size is partly dispersion. Section D's interpretation does not mention it, although the p-value is printed in the summary table.

Fix: report PERMDISP alongside every PERMANOVA in the narrative, and interpret wood as *more heterogeneous*, which is itself an ecologically meaningful finding.

P8 – MODERATE: ITS-level ordinations are degenerate (*already addressed*)

59% of sampling-unit pairs share no genotype, so their Bray–Curtis dissimilarity is tied at exactly 1. NMDS's default weak-tie handling leaves those pairs unconstrained, producing a near-zero stress that is an illusion; refitted with strong ties the stress is **0.249** (unreliable) at ITS level but **0.163** at genus level. The pipeline now reports both and labels the unusable plots.

P9 – MODERATE: genotype names are not clean OTUs

Taxon identity is a BLAST-derived **name string**, not a clustered unit. **86 of 284 labels (30%)** are ambiguous — e.g. “*Aspergillus brunneus* or *A. niveoglaucus* etc (4 spp.)”, “*Cladosporium subuliforme* or *C. xylophilum* etc (26 spp.) gen 2”. Some bundle dozens of species into one “taxon” while nearly identical sequences may sit under two labels.

The two input files also disagree on **18 names in each direction**, including plain typos (*guandongensis* / *guangdongensis*; *prosopidis* / *Prosopidis*; “*P cf. sporulosa*”). The lineage join silently falls back to matching on the leading genus word for these.

All 650 ITS sequences are present in sheet Feui12 (306–980 bp, one per isolate, 100% coverage) and are never used.

Fix: cluster the sequences into OTUs at 97% (or use ASVs), and use those as the analytical unit. This is the highest-value single change available — it removes the naming problem, and will very likely reduce the singleton fraction that is driving P2, P5 and P8.

P10 – MODERATE: zone analyses are largely untestable

Ficus wood zones have **1 sampling unit for four of six zones** (zone 1: 1, zone 2: 1, zone 3: 1, zone 4: 2, zone 5: 1, zone 6: 7). “No zone effect” here means “no test was possible”, not “no effect exists”. The same applies to the per-substrate orientation tests (1–4 units per bearing).

Fix: state the replication explicitly wherever a null result is reported; consider dropping the discrete-zone analysis in favour of the trunk-vs-branch contrast, which does have replication.

P11 — MINOR: no global multiplicity control

The pipeline generates roughly 19 PERMANOVA, 17 ANOSIM and 17 PERMDISP tests plus 120 pairwise rows. Holm correction is applied *within* each pairwise family only. At $\alpha = 0.05$ across ~50 top-level tests, 2–3 false positives are expected.

Fix: designate a small set of pre-specified primary hypotheses; label everything else exploratory.

3. Proposed methodology

Step	Current	Proposed
Analytical unit	BLAST name string	97% OTUs clustered from Feuil2 sequences (VSEARCH/DADA2)
Richness	Observed S only	Chao1/ACE + Good's coverage , reported with SEs
Diversity comparison	Indices on pooled data, $n = 1$	Hill numbers ($q = 0,1,2$) per sampling unit , coverage-standardised (iNEXT), tested across substrates
Rarefaction	Individual-based on pooled group	Sample-based rarefaction + extrapolation with CIs
Dissimilarity	Bray–Curtis on raw counts	Bray–Curtis on coverage-standardised data; add Raup–Crick (null-model based), which is designed for exactly this sparsity
Ordination	NMDS at ITS level	NMDS at genus/family level , or PCoA; report tie-aware stress
Model	<code>community ~ factor</code>	<code>community ~ depth + factor</code> , with tree/zone as stratum where replication allows
Indicators	raw $p \leq 0.05$	BH-FDR $q \leq 0.05$
Dispersion	computed, not discussed	reported next to every PERMANOVA

Priority order: P1 and P2 before anything is shown to anyone (both are wrong-as-published). Then P9 (OTU clustering), which improves P5 and P8 at the same time. Then P4 and P5, which are what the diversity question actually asks for. P3 is a framing fix, not an analysis fix, but it changes what the paper can claim.

4. Questions for the data owner

1. **How many trees were sampled per host?** If more than one, the tree ID needs adding — it determines whether the host comparison is publishable as a species-level result.
2. **Is “Lauraceae” leaves or wood?** LOT2_pooled_counts.xlsx labels the column “Lauraceae leaves”, but the samples workbook sheet is titled “66. Fungi-Endo wood (Host)”, and the 219 Lauraceae isolates match **none** of the three per-substrate sheets. The report’s central claim — that leaf-vs-wood tissue type is the strongest split — depends on this.
3. **Is the Ficus growing on the Lauraceae?** “Host” in the sheet title suggests a strangler relationship. If so the two trees are not independent, which matters for P3.

4. **What do the per-substrate sheets represent** (264 / 167 / 20 isolates) relative to the 650 analysed? Sheet 66 in particular contains only 20 isolates that appear nowhere in the analysis.
5. **Should isolates be de-replicated per sampling unit?** Several units contain the same genotype repeatedly; whether these are independent colonisations or one colony subsampled changes the abundance weighting throughout.

5. Resolution status

All eleven problems have been addressed in the pipeline, and the five provenance questions were answered by the data owner. Their answers are now recorded in the README ("Sampling design: what was sampled, and the assumptions behind it") so collaborators can check the reasoning.

#	Problem	Status
P1	Totals row plotted as a taxon	Fixed — <code>parse_L0T2.R</code> drops rows with neither culture code nor taxon name; pie/abundance charts now show the true 0–20% <i>Incertae sedis</i> , and an integrity check asserts pooled = sample-level totals
P2	Indicator species uncorrected	Fixed — Benjamini–Hochberg applied; full p/q tables written to <code>*_indicator_species_all.csv</code>
P3	Pseudoreplication	Documented — confirmed 1 tree per host, physically interlocked; <code>tree_id</code> added, scope caveat now heads the Conclusion
P4	Alpha diversity untestable	Fixed — <code>diversity_analysis.R</code> computes per-unit diversity with Kruskal–Wallis and pairwise Wilcoxon
P5	Completeness unquantified	Fixed — Chao1/ACE, Good’s coverage and equal-coverage richness in new Section A0
P6	Sampling depth confounded	Fixed — every PERMANOVA also reported adjusted for depth (marginal/type-III)
P7	Dispersion heterogeneity	Fixed — PERMDISP reported alongside PERMANOVA, and the wood-heterogeneity result is now interpreted rather than buried
P8	NMDS degeneracy	Fixed — tie-aware stress reported; OTU clustering removed the degeneracy at the primary level
P9	Name strings used as OTUs	Fixed — <code>cluster_otus.sh</code> clusters the 650 ITS sequences; 97% OTUs are now the analytical unit
P10	Zone analyses untestable	Documented — replication and an explicit caution printed in every PERMANOVA report
P11	No global multiplicity control	Partly fixed — FDR within IndVal, Holm across alpha indices and pairwise tests. Designating primary vs exploratory hypotheses remains an authorial decision

What the fixes changed in the results

- **OTU clustering rescued real signal.** On the raw name labels, *no* indicator taxon survived FDR correction. On 97% OTUs, **14 substrate indicators survive** — fewer tests and consolidated singletons recovered a result the old pipeline could not have reported honestly.
- **Alpha diversity is now testable and significant.** Richness and Shannon differ across substrates (Holm-adjusted $p = 0.0015$), while **evenness does not** ($p = 0.27$).
- **An apparent contradiction turned out to be a finding.** Per sampling unit, *Ficus* wood is the *least* diverse substrate; at equal coverage it is the *most* diverse. Leaf units are individually richer, but the wood community is more heterogeneous between units and has a longer tail of rare taxa — the same heterogeneity PERMDISP detects.
- **The substrate effect held up.** It survives adjustment for sampling depth, so it is not an artefact of unequal recovery.
- **One analysis was deleted, not fixed.** `lauraceae_leaves_trunk_vs_branch_*` compared branch against branch, because the Lauraceae trunk was never sampled.

Still open

Whether the Lauraceae material is leaves or branch wood. This is the one unresolved question, and the report's tissue-type conclusion depends on it. If it is branch wood, "the two leaf substrates resemble each other more than either resembles wood" cannot stand. Everything else is unaffected.

6. Addendum (2026-09-29): sampling-unit corrections

After this review, the data curator checked the sampling design with the field team (LoT) and found two labelling errors in `LOT2_samples.xlsx`. `parse_LOT2.R` now corrects both when it reads the workbook:

1. **All Lauraceae branches are in zone 6.** 53 isolates (S7, most of S8, one S4 isolate) were recorded as zone 5. Because a sampling unit is `substrate × zone × unit`, S4 and S8 were each split in two, so the Lauraceae had **10 units instead of 8**. One of the extra units held a single isolate and another held three.
2. **Ficus trunk wood is numbered S11–S15** (zone 5 down to zone 1), renamed from S9–S13 so it cannot be confused with canopy branches S9–S10, where only leaves were sampled. Three zone-4 isolates still carried the old label S10, which split zone 4 into two units.

The pipeline also now checks every sampling unit against the curator's design. The data go from **33 to 30 sampling units** (Ficus leaves 10, Ficus wood 12, Lauraceae 8). Figures quoted in sections 1–5 predate this correction. What changed:

Item	Before	After
P3: units treated as host replicates (leaf contrast)	20	18
P4: Lauraceae mean richness per unit	10.5 ± 5.5	12.9 ± 3.5
P4: Lauraceae vs Ficus wood, richness / Shannon (Holm p)	0.16 / 0.26	0.029 / 0.047
P6: depth, adjusted for substrate	$p = 0.006$	$p = 0.054$
P7: PERMDISP across substrates	$F = 7.2$	$F = 21.7$
Substrate PERMANOVA (97% OTUs)	$R^2 = 0.28$	$R^2 = 0.33$
P10: units per Ficus trunk zone	1–2	exactly 1
Substrate indicators (FDR $q \leq 0.05$)	14	13

Readings that changed.

- **Alpha diversity.** The two tiny Lauraceae units had pulled its per-unit mean down. The Lauraceae now separates from Ficus wood, not only Ficus leaves from wood.
- **Orientation within the Lauraceae** becomes nominally significant ($p = 0.048$). It does not survive Holm correction across the three substrates ($p = 0.144$), and it compares 4 north-facing with 4 north-west-facing branches. The pooled orientation conclusions are unchanged.
- **Zones.** With one unit per trunk zone, P10 applies with full force. The zone sweep is significant at class level only ($p = 0.017$; it was already $p = 0.025$ before the correction). The previous README conclusion that zones were non-significant “at any rank” did not match its own table, and now does.
- **ITS-label NMDS.** The ~ 0.0001 stress in the substrate ordination at name-label resolution came from the one-isolate unit, which shared no genotype with any other unit. With the unit merged, that ordination is no longer degenerate. It is still unreliable (tie-aware stress 0.20).

A reading that held, with a caveat on how to test it. The tissue-type conclusion still stands: by distance between substrate centroids, the two leaf substrates are the closest pair at every rank (new table `substrate_pairwise_multilevel.csv`, README Section D1). Pairwise PERMANOVA R^2 would suggest otherwise at some ranks. R^2 falls when a group is heterogeneous, and after the correction the Lauraceae units are far more homogeneous than the Ficus wood units, so R^2 is not a valid way to rank which substrates are most alike.

Newly open. The design lists 13 Ficus wood units (8 branches + 5 trunk zones), but **no isolates are recorded for branch S4 wood**, in either the main sheet or the per-substrate sheet. If S4 wood was sampled and yielded nothing, 12 is correct. If its isolates exist but are mislabelled, they are currently counted under another unit.